

Module 1 Assignment



Project #1: Password Verifier (50 points)

Objective:

The objective of this project is to create a Java program that verifies a password based on the following criteria:

1. The password length should be at least eight characters long.
2. The password must have at least one uppercase letter and one lowercase letter.
3. The password must have at least one digit.

At the end of this project, you should be able to:

- Create a Java program that validates user-input passwords.
- Apply conditional statements to check password criteria.
- Prompt the user for input and provide feedback on password validity.

Instructions:

Part 1: Setting Up Your Development Environment

1. Ensure you have Java Development Kit (JDK) installed on your computer.
2. Use a code editor or integrated development environment (IDE) of your choice (e.g., Eclipse, jGrasp, or Visual Studio Code).

Part 2: Creating the Java Program

In this part, you will create a Java program that validates user-entered passwords according to the specified criteria.

1. Create a new Java class named `PasswordVerifier`.
2. Write the Java code following the instructions

3. Save the Java file and submit it with a screenshot to Canvas.

Part 3: Testing the Program

1. Compile and run the `PasswordVerifier` program.
2. Follow the on-screen prompts to enter various passwords to test the validation criteria.
3. Observe the program's output, which should provide feedback on whether the entered passwords are valid or not.

Part 4: Experiment and Challenge (Optional)

1. Try modifying the program to accept multiple password attempts until a valid password is entered.
2. Experiment with different approaches to improve the user experience and the accuracy of password validation.

Conclusion:

In this lab project, you have successfully created a Java program that validates user-entered passwords according to specific criteria. You learned how to apply conditional statements and provide feedback to the user based on the validity of the password. This practical exercise is essential for understanding and implementing basic password validation in software applications.

Project #2 Weekly Sales Analysis (50 points)

Project Description:

This Java project aims to develop a program that reads a text file containing weekly sales data separated by commas. The numbers in the file are floating-point values. The program calculates various statistics, including the total sales for each week, the average daily sales for each week, the total of all weeks, the average weekly sales, and identifies the week numbers with the highest and lowest sales.

Project Requirements:

1. **File Input:** The program should read a text file (`sales_data.txt`) that contains weekly sales data, where each line represents one week. The values are separated by commas.
2. **Data Processing:** For each week:
 - Calculate the total sales for that week.
 - Calculate the average daily sales for that week.
 - Identify and record the week number with the highest sales.
 - Identify and record the week number with the lowest sales.
3. **Overall Statistics:** After processing all weeks:
 - Calculate the total sales for all weeks.
 - Calculate the average weekly sales (total sales divided by the number of weeks).
4. **Display Results:** Display the following results:
 - Total sales for each week.
 - Average daily sales for each week.
 - Total sales for all weeks.
 - Average weekly sales.
 - Week number with the highest sales.
 - Week number with the lowest sales.

Implementation Guidelines:

1. Create a Java program with appropriate classes and methods to handle file input, data processing, and result display.
2. Use appropriate data structures (e.g., arrays, lists) to store and manipulate the sales data.
3. Utilize file handling to read the weekly sales data from the text file.
4. Implement the required calculations for total sales, average daily sales, total of all weeks, and average weekly sales.
5. Track the week numbers with the highest and lowest sales as you process the data.

6. Display the calculated results clearly to the user.
7. Test the program with various input files and validate the results.

Project Objective:

The objective of this lab is to develop a Java program that reads a text file containing weekly sales data separated by commas. The program calculates various statistics, including the total sales for each week, the average daily sales for each week, the total of all weeks, the average weekly sales, and identifies the week numbers with the highest and lowest sales.

Instructions:

Part 1: Setting Up Your Development Environment

1. Ensure you have Java Development Kit (JDK) installed on your computer.
2. Use a code editor or integrated development environment (IDE) of your choice (e.g., Eclipse, IntelliJ IDEA, or Visual Studio Code).

Part 2: Creating the Java Program

In this part, you will create a Java program that reads a text file containing weekly sales data and calculates the specified statistics.

1. Create a new Java class named `WeeklySalesAnalyzer`.
2. Write the Java code
3. **Save the Java file and submit it with a screenshot to Canvas.**

Part 3: Testing the Program

1. Create a text file named `sales_data.txt` and enter sample sales data for multiple weeks, separated by commas.

Example `sales_data.txt` content:

Copy code

```
100.50, 150.25, 75.75, 120.80, 95.60 200.75, 180.90, 140.30, 165.50, 210.40 90.25, 110.30, 75.60, 135.90, 105.70
```

- 2. Compile and run the `WeeklySalesAnalyzer` program.
- 3. Observe the program's output, which should display the calculated statistics for each week, total sales for all weeks, average weekly sales, and the week numbers with the highest and lowest sales.

Part 4: Experiment and Challenge (Optional)

- 1. Modify the program to allow the user to input the file name and path.
- 2. Experiment with different input files containing various sales data.
- 3. Explore ways to enhance the program's error handling and user experience.

Conclusion:

In this project lab, you have successfully created a Java program that reads a text file containing weekly sales data, processes the data to calculate statistics, and displays the results. This practical exercise allows you to apply file handling and data processing concepts in a real-world context, which is valuable for various data analysis tasks.

Points 100

Submitting a file upload

Due	For	Available from	Until
Sep 17	Everyone	-	Oct 22 at 11:59pm